

Location Feature Setup Guide

☒ Working Solution

What This Does

When the user taps "**Get Current Location**":

1. Requests location permission (if not granted)
2. **Automatically shows Google Play Services "Turn on location?" dialog** if location is OFF
3. Fetches current coordinates with high accuracy
4. Updates latitude & longitude in the form

Key Feature

Built-in Location Dialog: Using `@react-native-community/geolocation` with:

- `forceRequestLocation: true` - Forces location request on Android
- `showLocationDialog: true` - **Shows Google Play Services popup to enable location**
- `forceLocationManager: false` - Uses Google Play Services (FusedLocationProvider)

This approach works natively without additional packages!

☒ Configuration Complete

Packages Used

- ☒ `@react-native-community/geolocation` - For getting device location (has built-in location enable dialog!)
- ☒ `react-native-permissions` - For runtime permission handling

Android Configuration

1. **AndroidManifest.xml** ☒

Location: `android/app/src/main/AndroidManifest.xml`

Already configured with:

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_BACKGROUND_LOCATION" />

<meta-data android:name="com.google.android.gms.version"
            android:value="@integer/google_play_services_version" />
```

2. **build.gradle** ☒

Location: `android/app/build.gradle`

Already configured with:

```
implementation 'com.google.android.gms:play-services-location:21.0.1'
```

How It Works

User Flow

1. User taps "**Get Current Location**" button
2. App requests **location permission** (if not granted)
3. `Geolocation.getCurrentPosition()` is called with `showLocationDialog: true`
4. **Google Play Services automatically shows "Turn on location?" dialog** (if location is OFF)
5. User taps "**Turn on**" → Location services enabled
6. App fetches **current coordinates** (high accuracy)
7. **Latitude & Longitude** populated in form fields

Code Implementation

```
Geolocation.getCurrentPosition(  
  successCallback,  
  errorCallback,  
  {  
    enableHighAccuracy: true,  
    timeout: 15000,  
    maximumAge: 10000,  
    forceRequestLocation: true, // Forces location request  
    forceLocationManager: false, // Use Google Play Services  
    showLocationDialog: true, // 📍 Shows "Turn on location?" dialog!  
    distanceFilter: 0,  
  }  
);
```

Features

- ✓ **Built-in location dialog** - No extra packages needed!
- ✓ **Crash-proof** - All calls wrapped in try/catch
- ✓ **User-friendly errors** - Specific messages for each error type
- ✓ **Settings deeplink** - "Open Settings" button when needed
- ✓ **High accuracy** - Uses Google Play Services Fused Location
- ✓ **Console logging** - Detailed logs with `[Location]` prefix

Testing

Test 1: Location OFF

1. Turn off device location in Settings
2. Open app → Edit Profile → Tap "Get Current Location"
3. **Expected:** Permission dialog → Grant permission
4. **Expected:** Google Play Services dialog: "Turn on location?"
5. Tap "Turn on"
6. **Expected:** Location fetched and displayed

Test 2: Location ON

1. Keep device location enabled
2. Tap "Get Current Location"
3. **Expected:** Location fetched immediately (no dialog)

Test 3: Permission Denied

1. Deny location permission
2. **Expected:** Alert with "Open Settings" button

Why This Works Better

Previous Attempt ([react-native-location-enabler](#))

- ✗ Outdated package with compatibility issues
- ✗ Build errors: "Unresolved reference 'currentActivity'"
- ✗ Not compatible with React Native 0.76+
- ✗ Extra native module to maintain

Current Solution ([@react-native-community/geolocation](#))

- ☒ Official React Native Community package
- ☒ Well-maintained and actively updated
- ☒ Built-in location enable dialog via [showLocationDialog](#)
- ☒ Uses Google Play Services Fused Location
- ☒ No additional native linking required
- ☒ Works on all Android versions (5.0+)

Geolocation Options Explained

Option	Value	Purpose
enableHighAccuracy	<code>true</code>	Use GPS for accurate coordinates
timeout	<code>15000</code>	Max time (15s) to wait for location
maximumAge	<code>10000</code>	Max age (10s) of cached location
forceRequestLocation	<code>true</code>	Force location request on Android
forceLocationManager	<code>false</code>	Use Google Play Services (better)
showLocationDialog	<code>true</code>	 Show "Turn on location?" dialog

Option	Value	Purpose
<code>distanceFilter</code>	<code>0</code>	No distance filter for fresh location

Build Commands

```
# If you need to rebuild
npm run android
```

Files Modified

1. ☒ `src/authencation/user/EditProfileScreen.tsx` - Location implementation
2. ☒ `android/app/src/main/AndroidManifest.xml` - Permissions (already configured)
3. ☒ `android/app/build.gradle` - Play Services (already configured)

Status: ☒ Build successful! App ready to test.

The "Turn on location?" dialog will appear automatically when:

- Location permission is granted
- Device location is turned OFF
- User taps "Get Current Location"

This is handled natively by `@react-native-community/geolocation` with the `showLocationDialog: true` option!